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Lecture 7: Instrumental variables estimation

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Why we like experiments

- Identification comes for free: we can ensure that treatment assignment is independent of potential outcomes.
- Methods for analysis and inference are (relatively) straightforward.
- However, treatment assignment doesn't guarantee treatment uptake. (examples: GOTV, de-worming; ethical considerations)
- So we may have independence of treatment assignment and potential outcomes, but not *actual* treatment received and potential outcomes.

Noncompliance

- People who opt-out of treatment will often look different, and have different associated experimental responses, as compared to people who will opt-in when treatment is offered.
- We can focus just on *intention to treat* (ITT) effects: what is the effect on a group of having treatment offered to them *whether they take up treatment or not*.
- Combination of effects among people who *do* take up treatment when randomized to receive it, and (potentially null) effects among people who *don't*. (We'll come back to compliance types).
- Often these are of policy relevance—for things you can't *force* people to take, it's worth understanding what the average effect of making them available is. And identification is still good.
- However, this might not be a quantity that we're interested in, and when the assignment mechanism would look different in the "real" world, external validity may be poor.

Instrument

- With noncompliance when we can't *force* everyone to take treatment, we can consider effects on the subgroup for whom random assignment was effective at moving them to uptake treatment.
- Our estimand becomes the "Local Average Treatment Effect" or "Complier Average Treatment Effect."

Observational data motivation

- Motivation from natural experiments (Angrist and Krueger, 2001)
- Uses only a part of the variation in the treatment received
- Relevance of estimand? Do we think there is **heterogeneity** in response?
- In observational setting, what is the alternative?

Instrument

- Call the treatment *assignment* an *instrument* $Z_i \in \{0, 1\}$
- Call the treatment *received* W_i;
  - Treatment *received* is a function of treatment *assigned*: $W_i(Z_i) \in \{0, 1\}$
- We can denote outcomes as a function of treatment assigned and received: $Y_i^{obs} = Y_i(Z_i, W_i(Z_i))$

Compliance behavior

$$C_i = \begin{cases} \text{complier} & W_i(0) = 0, W_i(1) = 1, \\ \text{defier} & W_i(0) = 1, W_i(1) = 0, \\ \text{always-taker} & W_i(0) = 1, W_i(1) = 1, \\ \text{never-taker} & W_i(0) = 0, W_i(1) = 0 \end{cases}$$

IV assumptions

- No defiers (monotonicity): $W_i(1) \geq W_i(0)$ (is this plausible?)
- Exclusion restriction: $Y_i(z, w) = Y_i(z', w),$ for all z, z', w (is this plausible?)

$$\Rightarrow Y_i(Z_i, W_i) = Y_i(W_i)$$

ATE

$$\tau^{ate} = \mathrm{E}\left[Y_i(1) - Y_i(0)\right]$$

But

$$\mathrm{E}\left[Y_i(1)\right] \neq \mathrm{E}\left[Y_i^{obs} | Z_i = 1\right] \neq \mathrm{E}\left[Y_i^{obs} | W_i = 1\right]$$
$$\mathrm{E}\left[Y_i(0)\right] \neq \mathrm{E}\left[Y_i^{obs} | Z_i = 0\right] \neq \mathrm{E}\left[Y_i^{obs} | W_i = 0\right]$$

With noncompliance, ATE is not identified without further assumptions.

ITT

$$
\begin{aligned}
\tau^{itt} &= \mathrm{E}\left[Y_i(1, W_i(1))\right] - \mathrm{E}\left[Y_i(0, W_i(0))\right] \\
&= \mathrm{E}\left[Y_i(W_i)|Z_i = 1\right] - \mathrm{E}\left[Y_i(W_i)|Z_i = 0\right] \\
&= \mathrm{E}\left[Y_i^{obs}|Z_i = 1\right] - \mathrm{E}\left[Y_i^{obs}|Z_i = 0\right]
\end{aligned}
$$

With noncompliance, ITT *is* identified.
But it does not necessarily correspond to ATE.

LATE

$$\begin{aligned}\tau^{late} &= \mathrm{E}\left[Y_i(1) - Y_i(0) | C_i = \text{complier}\right] \\ &= \frac{\mathrm{E}\left[Y_i^{obs} | Z_i = 1\right] - \mathrm{E}\left[Y_i^{obs} | Z_i = 0\right]}{\mathrm{E}\left[W_i^{obs} | Z_i = 1\right] - \mathrm{E}\left[W_i^{obs} | Z_i = 0\right]}\end{aligned}$$

LATE also does not necessarily correspond to ATE! Also different from ITT.

Estimating methods

Fabricated data: true treatment effect is 2

```
> library(fabricatr)
> library(estimatr)
> set.seed(60637)
> dat ← fabricate(
+ N = 1e4,
+ Z = rbinom(N, 1, prob = 0.4),
+ W = Z * rbinom(N, 1, prob = 0.8),
+ Y = rpois(N, lambda = 4) + W * rpois(N, lambda = 2),
+ X = rnorm(N)
+ )
```

Estimating methods

```
> head(dat)
```

```
     ID Z W  Y          X
1 00001 1 1 11 -0.7181921
2 00002 1 0  0  0.1173308
3 00003 1 1 11 -0.9540193
4 00004 1 1  6 -0.2415993
5 00005 0 0  5  1.0741809
6 00006 0 0  4  0.7435195
```

Estimating methods: ITT

Difference in means estimate

```
> mean(dat$Y[which(dat$Z == 1)]) -
+ mean(dat$Y[which(dat$Z == 0)])
```

```
[1] 1.602577
```

Estimating methods: LATE

- Generically:

$$\hat{\tau}^{late} = \widehat{\mathrm{Cov}}\left[Y_i, Z_i\right] / \widehat{\mathrm{Cov}}\left[W_i, Z_i\right]$$

- Wald estimator: With binary instrument, equivalently:

$$\hat{\tau}^{late} = \frac{\hat{\mathrm{E}}\left[Y_i^{obs} | Z_i = 1\right] - \hat{\mathrm{E}}\left[Y_i^{obs} | Z_i = 0\right]}{\hat{\mathrm{E}}\left[W_i^{obs} | Z_i = 1\right] - \hat{\mathrm{E}}\left[W_i^{obs} | Z_i = 0\right]}$$

Estimating methods: LATE

```
> cov(dat$Y, dat$Z)/cov(dat$W, dat$Z)
```

```
[1] 2.013196
```

Estimating methods: LATE (continued)

```
> (numerator ← mean(dat$Y[which(dat$Z == 1)]) -
+ mean(dat$Y[which(dat$Z == 0)]))
```

```
[1] 1.602577
```

```
> (denominator ← mean(dat$W[which(dat$Z == 1)]) -
+ mean(dat$W[which(dat$Z == 0)]))
```

```
[1] 0.7960361
```

```
> numerator/denominator
```

```
[1] 2.013196
```

Estimating methods: LATE

- Two-stage least squares: regress treatment realized on instrument (and other instruments, covariates); regress outcome on predicted value of treatment (and control covariates)

Estimating methods: LATE

Doing each stage separately:

```
> dat$W_predicted ← lm_robust(W ~ Z, data = dat)$
    fitted.values
> lm_robust(Y ~ W_predicted, data = dat)
```

```
             Estimate Std. Error   t value      Pr(>|t|) CI Lower
(Intercept) 4.008813 0.02590565 154.74667  0.000000e+00 3.958032
W_predicted 2.013196 0.05949577  33.83764 1.001255e-237 1.896573
             CI Upper   DF
(Intercept) 4.059593 9998
W_predicted 2.129820 9998
```

Estimating methods: LATE (continued)

With `estimatr`:

```
> iv_robust(Y ~ W| Z, data = dat)
```

```
            Estimate Std. Error  t value      Pr(>|t|) CI Lower
(Intercept) 4.008813 0.02590565 154.7467  0.000000e+00 3.958032
W           2.013196 0.05713043  35.2386 1.627323e-256 1.901209
            CI Upper   DF
(Intercept) 4.059593 9998
W           2.125184 9998
```

Estimating methods: LATE

With covariates:

```
> iv_robust(Y ~ W + X | Z + X, data = dat)
```

```
             Estimate Std. Error   t value      Pr(>|t|)
(Intercept)  4.00873400 0.02590631 154.73967  0.000000e+00
W            2.01258736 0.05712699  35.23006 2.134297e-256
X           -0.02730166 0.02131311  -1.28098  2.002305e-01
             CI Lower   CI Upper   DF
(Intercept)  3.95795242 4.05951558 9997
W            1.90060696 2.12456776 9997
X           -0.06907965 0.01447632 9997
```

Estimating methods: LATE

- TSLS allows for multiple instruments, covariates
- consistent but *not* unbiased

Possible issues with instruments

- Failure of exogeneity (examples in the experimental setting?)
- Weak instruments
  - Diagnose strength using the first stage, $W \sim Z$ (plus any covariates): how strongly does the instrument predict treatment received? Use an appropriate first-stage F statistic.
  - The reduced form, $Y \sim Z$, estimates the ITT effect; a small effect need not imply a weak instrument.

Alternatives

- Assume unconfoundedness

$$W_i \perp\!\!\!\perp (Y_i(1), Y_i(0)) | X_i$$

- Estimate average effect for compliers, generalize to population
- Test equalities of compliers under treatment, always takers; compliers under control, never-takers; following Athey and Imbens (2017)

Alternatives

- Sometimes we can also *bound* treatment effects.
- Manski bounds: worst case bounds

Take-aways for experiments

- What is the quantity of interest in your experimental setting?
- When would you care about ATE? ITT? LATE?
- Is non-compliance likely to be a problem? (How do you define *compliance*?)
- Do you expect compliers and non-compliers to look different?
- Do you expect compliers and non-compliers to have different (hypothetical) treatment effects?

References I

Angrist, J. D. and Krueger, A. B. (2001). Instrumental variables and the search for identification: From supply and demand to natural experiments. Journal of Economic perspectives, 15(4):69–85.

Athey, S. and Imbens, G. W. (2017). The econometrics of randomized experiments. In Handbook of economic field experiments, volume 1, pages 73–140. Elsevier.